## 1 MEASURE

fingerprint: neighbouring-pixel correlation
on the raw inputs (seconds of CPU, no training)

near-zero correlation = encoded data
high correlation = image-like data

tight budget? under 35 min skip stages 2, 4, 5;
under 15 min also skip stage 3

## 2 BACKBONE CHOICE

image-like -> 2-D TCN directly, no bake-off
(channel-mixing stem if 8 or more channels)

encoded -> gate: train CNN, 1-D TCN and 2-D TCN
briefly at matched wall-clock

adopt the leader only if it clearly beats the CNN,
otherwise default to the CNN

## 3 AUGMENTATION PROBE

two small networks of the chosen family,
trained briefly: with vs without augmentation

augmentation off (plus mixup) only when OFF wins
by a clear margin: it is regularisation, OFF leads early

otherwise: augmentation stays on (the default)

## 4 DEPTH SEARCH

depth-4 and depth-8 candidates trained briefly
against the incumbent default depth 6

adopt a candidate only if it beats the incumbent
beyond noise AND survives re-training verification

otherwise: keep depth 6
(skipped for the 1-D TCN)

## 5 RANDOM SEARCH ARM

runs only on encoded data, only if budget remains

sample 6 random architectures
(11 operation types, depth 3 to 12, width 16 to 48)

adopt only if clearly better than the incumbent,
otherwise discard

## 6 TRAIN TO BUDGET

the winner trains on ALL remaining time, cosine
learning rate annealed to zero exactly at the budget

keep the best checkpoint -> predictions